

COMPILER DESIGN

ASSIGNMENT

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Course Code : CSA1405

Course Name : Compiler Design

Slot : C

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①

1) Error analysis.

a) Error Identification.

<u>Line</u>	<u>Error</u>	<u>Description.</u>
6	int num = 10;	Identifier cannot digit.
7	float b = 5.5	Missing ;
9	c - add (10);	function called
10	if (c == 20)	Assignment operator
11	printf ("value is 20\n")	Missing ;
12	d = c + 5;	variable d not declared
16	* p = 100;	p not initialized
18	x / y	Division by zero.
19-21	while (x < 15)	Infinite loop.

②

b) lexical Error

- * num

syntax Error

- * float b = 5.5

- * printf ('')

semantic Error

- * wrong number of
function

- * undeclared variable

logical Error

- * if (C == 20)

- * Infinite loop.

c) compilation Behaviors

- * Invalid identifiers

- * Missing semicolons

- * undeclared variable

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d) corrected program.

```
#include <stdio.h>
int add (int a, int b)
{
    return a+b;
}
int main()
{
    int num = 10;
    float b = 5.5;
    int c;
    c = add(10, 10);
    if (c == 20)
        printf("value is 20\n");
    int d = c + 5;
    int arr[5] = {1, 2, 3, 4, 5};
    printf("%d\n", arr[4]);
    int value = 100;
    int *p = &value;
    *p = 100;
    int x = 10, y = 2;
    int z = x / y;
    return 0;
}
```

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c) token connection

<u>Invalid</u>	<u>Correct</u>
value	value2
rate@	rate
'ab'	'a'
co\$5	105
num#1	num1
"Hello world"	"Hello world"
09	9

d) Advanced Thinking

- first → letters or _
- Remaining → letters,
digit or _

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e) Specifying Input

* scanf()

- %d → integer

- %f → float

- %c → character

- %s → string

* getch() ; getche()

- single character input

* gets()

- string input

* getw()

- integer input

f) Constants

* Types → integer, float,
character, string

* Rules → by user

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g) Defining Rules

```
# define PI 3.14159
```

```
# define MAX(a,b) ((a)>(b) ? (a) : (b))
```

```
void area (int r)
```

```
{
```

```
    float result ;
```

```
    result = PI * r * r ;
```

```
    printf ("area is %f", result) ;
```

```
}
```

h) Compilation

- * Save the file with extension .c

- * Compile → gcc file.c

- * run → ./a.out

i) Style and Convention

- * Use meaningful names

- * Indentation

- * Comments (// or /* */)

- * Keep lines short